Course: Linear Algebra and Optimization **Course Code: 23MA4BSLAO**

Unit 3: Inner Product Spaces

I. Inner product spaces, angle between vectors, Length and Orthogonality:

1. Verify that $\mathbb{R}^n$ with $\langle\ ,\ \rangle$ defined as the dot product is an inner product space. Also Expand $\langle 5u_1+8u_2, 6v_1-7u_2\rangle$ and $\|2u-3v\|^2$.
2. If $u=(1,3,-4,2), v=(4,-2,2,1)$ and $w=(5,-1,-2,6)$ in $\mathbb{R}^4$ then verify that $\langle 3u-2v, w\rangle = 3\langle u,w\rangle - 2\langle v,w\rangle$
3. Let $V=M$ be a vector space of all matrices of size $m\times n$, then prove that $\langle A,B\rangle = tr\left(B^T A\right)$ is an inner product.
4. Let $V=\mathbb{R}^2$ be a vector space with $u=(u_1,u_2)$ and $v=(v_1,v_2)$. Then prove that $\langle u,v\rangle = u_1v_1-u_1v_2-u_2v_1+4u_2v_2$ is an inner product.
5. Consider $u=(1,2,4)$, $v=(2,-3,5)$, $w=(4,2,-3)$ in $\mathbb{R}^3$, find $u.v$, $u.w$, $v.w$, $(u+v).w$, $\|u\|$ and $\|v\|$.
6. Consider $f(t)=3t-5$, $g(t)=t^2$ in $P(t)$ then the inner product is defined as $\langle p,q\rangle = \int_0^1 p(t)q(t)dt$. Find $\langle f,g\rangle$, $\|f\|$ and $\|g\|$.
7. Consider $f(t)=t+2$, $g(t)=3t-2$ and $h(t)=t^2-2t-3$ in $P(t)$ with the inner product defined as $\langle p,q\rangle = \int_0^1 p(t)q(t)dt$. Find $\langle f,g\rangle$, $\langle f,h\rangle$, $\|f\|$ and $\|g\|$. Normalize f and g.
8. If $u=(1,3,-4,2), v=(4,-2,2,1)$ in $\mathbb{R}^4$ then prove the following:
   a) triangular inequality: $\|u+v\|\le\|u\|+\|v\|$,
   b) Schwartz inequality: $\langle u,v\rangle^2 \le \langle u,u\rangle\langle v,v\rangle$ or $|\langle u,v\rangle|\le\|u\|\|v\|$.
   c) Parallelogram law: $\|u+v\|^2+\|u-v\|^2 = 2\left(\|u\|^2+\|v\|^2\right)$
9. Find the angle between the vectors
   a) $u(2,3,5)$ and $v=(1,-4,3)$ in $\mathbb{R}^3$ w.r.t the usual inner product.
   b) $u=(1,3,-5,4)$ and $v=(2,-3,4,1)$ in $\mathbb{R}^4$.
   c) $A=\begin{bmatrix}9&8&7\\6&5&4\end{bmatrix}$ and $B=\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}$ where $\langle A,B\rangle = Tr\left(B^T A\right)$.
   d) $f(t)=3t-5$ and $g(t)=t^2$ in $P(t)$ with $\langle p,q\rangle = \int_0^1 p(t)q(t)dt$.
10. Check whether $f(t)=\sin(t)$ and $g(t)=\cos(t)$ in $C[a,b]$ with $\langle f,g\rangle = \int_{-\pi}^{\pi} f(t)g(t)dt$ are orthogonal.

11. Consider a matrix $A=\begin{bmatrix}1&2&1\\2&5&4\\1&4&6\end{bmatrix}$ and the vectors $u=\begin{bmatrix}1\\\alpha\\1\end{bmatrix}$ and $v=\begin{bmatrix}2\\2\\3\end{bmatrix}$. The inner product $\langle u,v\rangle = u^T Av$. Find the value of α if it is known that the vectors u and v are orthogonal and hence find the length of u.
12. Find a nonzero vector that is orthogonal to $(1,2,1)$ and $(2,5,4)$ in $\mathbb{R}^3$.

II. Orthogonal complements and orthogonal basis:

1. Let W be a subspace of $\mathbb{R}^5$ spanned by $u=(1,2,3,-1,2)$ and $v=(2,4,7,2,-1)$. Find a basis of $W^{\perp}$.
2. Find a basis of W of $\mathbb{R}^4$ orthogonal to $u_1=(1,-2,3,4)$ and $u_2=(3,-5,7,8)$.
3. Find an orthogonal basis for $w^{\perp}$ where $w=(1,2,3,1)$.
4. Find an orthogonal basis for $w^{\perp}$ where $w=(1,-2,-1,3)$.
5. Show that $S=\{u_1=(1,1,0,-1),u_2=(1,2,1,3),u_3=(1,1,-9,2),u_4=(16,-13,1,3)\}$ is orthogonal and a basis of $\mathbb{R}^4$. Find the coordinates of the arbitrary vector $v=(a,b,c,d)$ in $\mathbb{R}^4$ relative the basis S.
6. Show that $S=\{u_1=(1,1,1,1),u_2=(1,1,-1,-1),u_3=(1,-1,1,-1),u_4=(1,-1,-1,1)\}$ is orthogonal and a basis of $\mathbb{R}^4$. Express $v=(1,3,-5,6)$ as a linear combination of the vectors of S. Find the coordinates of the arbitrary vector $v=(a,b,c,d)$ in $\mathbb{R}^4$ relative the basis S.
7. $P_2(t)$ is the vector space of polynomials of degree ≤ 2 with $\langle f,\ g\rangle=\int_0^1 f(t)g(t)dt$. Find a basis of the subspace W orthogonal to $h(t)=2t+1$.

III. Projections.

1. Find the projection of the vector v along w or $W=span(w_i)$:
   a) $v=(1,-2,3,4)$ along $w=(1,2,1,2)$ in $\mathbb{R}^4$.
   b) $v=(1,3,1,2)$ along $w=(1,-2,7,4)$ in $\mathbb{R}^4$.
   c) $v=(1,2,3,4,6)$ along $W=span\{(1,2,1,2,1),(1,-1,2,-1,1)\}$ in $\mathbb{R}^5$.
   d) $v=(1,3,5,7)$ along $W=span\{(1,1,1,1),(1,-3,4,-2)\}$ in $\mathbb{R}^4$.
   e) $v=t^2$ along $w=t+3$ in $P(t)$ with $\langle f,\ g\rangle=\int_0^1 f(t)g(t)dt$.
   f) $v=\begin{bmatrix}1&2\\3&4\end{bmatrix}$ along $w=\begin{bmatrix}1&1\\5&5\end{bmatrix}$ in $M_{2\times 2}$ with $\langle A,B\rangle = tr(B^T A)$.

IV. Gram-Schmidt orthogonalization:

1. Find an orthogonal basis and hence an orthonormal basis of the subspace W spanned by the following vectors
   a) $v_1=(1,1,1,1)$, $v_2=(1,2,4,5)$ and $v_3=(1,-3,-4,-2)$ of $\mathbb{R}^4$.
   b) $v_1=(1,1,1,1)$, $v_2=(1,1,2,4)$ and $v_3=(1,2,-4,-3)$ of $\mathbb{R}^4$.

c) $v_1=(1,1,1,1)$, $v_2=(1,-1,2,2)$ and $v_3=(1,\ 2,\ -3,\ -4)$ of $\mathbb{R}^4$.

d) $S=\{1,\ t,\ t^2\}$ of $P_2(t)$ given $\langle f,g\rangle=\int_0^1 f(t)g(t)\,dt$.

e) $S=\{1,\ t,\ t^2,t^3\}$ of $P_3(t)$ given $\langle f,g\rangle=\int_{-1}^1 f(t)g(t)\,dt$.

2. Find an orthogonal matrix P whose first row is $u_1=\left(\frac{1}{3},\frac{2}{3},\frac{2}{3}\right)$.

3. Find an orthogonal matrix P whose first two rows are linear combination of $u_1=(1,1,1)$ and $u_2=(1,-2,3)$.

V. QR factorization:

1. Obtain the QR factorization of the following matrices:

a) $A=\begin{bmatrix}1 & -1 & -1\\ 1 & 0 & 0\\ 1 & -1 & 0\\ 0 & 1 & -1\end{bmatrix}$.

b) $A=\begin{bmatrix}2 & -1\\ -1 & 3\\ 0 & 1\end{bmatrix}$.

c) $A=\begin{bmatrix}1 & 0 & 2\\ -1 & 2 & 0\\ -1 & -2 & 2\end{bmatrix}$.

d) $A=\begin{bmatrix}1 & 0 & -1\\ 2 & -3 & 3\\ -1 & 2 & 4\end{bmatrix}$.

e) $A=\begin{bmatrix}1 & 2\\ -1 & -2\\ 1 & 1\end{bmatrix}$.

2. Solve the system of equations $AX=B$ where $A=\begin{bmatrix}2 & 3\\ 2 & 4\\ 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}7\\ 3\\ 1\end{bmatrix}$ by QR factorization method.

3. Solve the system of equations $AX=B$ where $A=\begin{bmatrix}3 & -2\\ 2 & -3\\ 1 & -1\\ 2 & 3\\ 3 & 4\end{bmatrix}$ and $B=\begin{bmatrix}2\\ -1\\ 0\\ 1\\ 0\end{bmatrix}$ by QR factorization method.

VI. Least square method:

1. Solve the following system of equations $AX=b$ by the method of least squares and hence find the least square error:

a) $A=\begin{bmatrix}2 & 1\\ 1 & 0\\ 0 & -1\\ -1 & 1\end{bmatrix}$ and $b=\begin{bmatrix}3\\ 1\\ 2\\ -1\end{bmatrix}$.

b) $A=\begin{bmatrix}3 & -2\\ 2 & -3\\ 1 & -1\\ 2 & 3\\ 3 & 4\end{bmatrix}$ and $b=\begin{bmatrix}2\\ -1\\ 0\\ 1\\ 0\end{bmatrix}$.

2. In an experiment designed to determine the extent of a person's natural orientation, a subject is put in a special room and kept there for a certain length of time. He is then asked to find a way of a maze and record is made of the time it takes the subject to accomplish this task. The following data are obtained.

Time in Room (hours)	1	2	3	4	5	6
Time to find Way out of maze(minutes)	0.8	2.1	2.6	2.0	3.1	3.3

Let x denote the number of hours in the room and let y denote the number of minutes that it takes the subject to find his way out.

a. Find the least squares line of the form $y=a+bx$.

b. Use the equation obtained in (a) to estimate the time it will take the subject to find his way out of the maze after 10 hours in the room.

3. Find the least squares line of the form $y=a+bx$ for the data $(0,-1)$, $(1,3)$, $(2,5)$ and $(3,7)$.

4. A steel producer gathers the following data.

Year	1997	1998	1999	2000	2001	2002
Annual Sales (millions of dollars)	1.2	2.3	3.2	3.6	3.8	5.1

Represent the years 1997,…2002 as 0,1,2,3,4,5 respectively and let x denote the year. Let y denote the annual sales (in millions of dollars).

a. Find the least squares line of the form $y=ax+b$.

b. Use the equation obtained in (a) to estimate the annual sales for the year 2006.

5. A sales organisation obtains the following data relating the number of salespersons to annual sales.

Number of salespersons	5	6	7	8	9	10
Annual Sales (millions of dollars)	2.3	3.2	4.1	5.0	6.1	7.2

Let x denote the number of salespersons and let y denote the annual sales (in millions of dollars).

a. Find the least squares line of the form $y=a+bx$.

b. Use the equation in (a) to estimate the annual sales when there are 14 salespersons.

6. Find the least square parabola of the form $y=a+bx+cx^2$ for the data (1, 1), (2, 4), (3, 7) and (4, 5).

7. Find the least square parabola of the form $y=a+bx+cx^2$ for the data (1, 1), (2, 1), (3, 2) and (4, 5).